Abdoulaye Chaibou Saidou

Engineering student · Final-year internship — AI Research / R&D

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PROFILE

Fifth-year engineering student (IATD-SI, ENSAM Meknès) seeking a final-year internship in AI research or R&D. Interested in reinforcement learning, explainability and rigorous model evaluation; I state the limits of my work (synthetic or simulated data, negative results). A PhD is considered in the longer term.

Fast, continuous learner: picked up Julia, Rust, Unity / AR, RL and MLOps through my projects — I never stop learning.

SKILLS

Research: reinforcement learning (SAC), explainability (occlusion, prototypes), anomaly detection, benchmark-based evaluation

ML: Python, scikit-learn, Gymnasium, stable-baselines3, Isolation Forest, trees / random forests

Languages: Python, Julia, SQL

Tools: Jupyter, TensorBoard, Streamlit, Git

EXPERIENCE

Full-Stack Developer Intern — DigiScia, Ouagadougou (Burkina Faso) Jul – Sep 2026

- SimpleOQuotidien project (connecting small-trade service providers with clients: search, quotes, assignment); my part: the front-end, in Next.js.

Data Scientist Intern (remote) — Technocolabs Softwares, Indore (India) Aug – Oct 2025

- Team project “Startup Acquisition Analysis & ML Pipeline”: predicting startup outcomes (acquisition, closure, IPO, continued operation).
- My part: decision trees and SVM on the “companies” dataset.
- Contributed to preprocessing, feature engineering and cross-validation (per the certificate).

PROJECTS

Portefeuille-Advisor — reinforcement learning Gymnasium, stable-baselines3, Streamlit, TensorBoard

Academic project, Jan 2026 · supervisor Prof. Tawfik Masrour

- Gymnasium environment (3 assets, synthetic market, 120 months) and Soft Actor-Critic agent; reward = log return − λ·volatility, 3 risk profiles.
- Evaluated over 20 episodes against 3 benchmarks: the agent does not outperform 60/40 (Sharpe ≈ 0.03 vs ≈ 0.77); negative result reported as is.

AVIEN — industrial predictive maintenance FastAPI, Celery, Redis, TimescaleDB, Next.js, Docker Compose

Academic project, June 2026 · supervisor Prof. Smail Zaki

- Isolation Forest on sensor data, linear RUL estimate with confidence interval, FFT vibration diagnostics and occlusion-based explainability.
- Limitations made explicit: simulated data, no quantitative evaluation for lack of labels.

Random Forest from scratch (Julia, Python, R) Julia

Course project, team of 5 · my part: Julia implementation

- Implemented a random forest (entropy splits, bootstrap, majority vote) in Julia; compared with the team’s Python and R versions on synthetic data.
- Result: Julia fastest (1.85 to 6.50 s), close accuracies (0.62–0.74 vs 0.64–0.75 for Python).

ProtoPNet — study of an interpretable prototype network Python, Jupyter

Course project (CUB-200-2011) · teaching work

- Study of the ProtoPNet model (bird classification via prototypes): explanatory notebook and presentation for classmates.
- No contribution to the model or architecture (researchers’ work).

AI-generated document detector NLP

Fifth-year project, ongoing · Prof. Hosni, Prof. Fihri

- Detection of reports (final-year theses, etc.) written or not by AI; first stage: surveying suitable models and architectures.

EDUCATION

Engineering degree — ENSAM Meknès, IATD-SI track 2022 – 2027 (degree expected 2027)

LANGUAGES

French: working language, fluent · English: fluent written, intermediate spoken